

Practice Set 2

MA628: Financial Derivatives Pricing

January 16, 2026

1. A company announces a cash dividend of Rs. 6 per share. The stock closes at Rs. 240 one day before the ex-dividend date.
 - (a) Calculate the theoretical ex-dividend price.
 - (b) If an investor holds 250 shares, compute:
 - Dividend received
 - Total portfolio value immediately after the stock goes ex-dividend
2. A company announces a bonus issue in the ratio 3 : 2 (3 bonus shares for every 2 held). Before the announcement:
 - Market price per share = Rs. 180
 - Number of outstanding shares = 40 lakh
 - (a) Calculate the new number of outstanding shares.
 - (b) Find the theoretical ex-bonus price.
 - (c) Check whether the market capitalization changes due to the bonus issue.
3. A stock trading at Rs. 1,200 undergoes a stock split in the ratio 1 : 5.
 - (a) Calculate the new face value if the original face value was Rs. 10.
 - (b) Find the post-split market price.
 - (c) If an investor initially owned 80 shares, compute:
 - New number of shares
 - Total investment value after the split
4. A company announces a rights issue of 1 share for every 4 shares held at a price of Rs.150. The current market price is Rs. 210.
 - (a) Compute the theoretical ex-rights price (TERP).
 - (b) Calculate the value of one right.
 - (c) If an investor owns 400 shares, determine:
 - Number of rights shares available
 - Total amount required to subscribe fully

5. A price-weighted index consists of the following stocks:

Stock	Price (Rs.)
A	120
B	80
C	200
D	100

- (a) Calculate the index value.
 (b) Stock C undergoes a 2 : 1 stock split. Compute the new divisor to keep the index unchanged.

6. An index consists of three stocks:

Stock	Price (Rs.)	Shares Outstanding (crore)
X	500	2
Y	250	4
Z	100	10

- (a) Compute the total market capitalization.
 (b) Find the weight of each stock in the index.
 (c) If stock X rises by 10%, estimate the approximate percentage change in the index.

7. The spot exchange rate is:

$$1 \text{ USD} = \text{Rs.}83.20$$

The 6-month forward rate is quoted at Rs. 84.00.

- (a) Determine whether the INR is at a forward premium or discount.
 (b) Calculate the annualized forward premium/discount percentage.

8. Given:

$$1 \text{ USD} = \text{Rs.}83.50, \quad 1 \text{ EUR} = 1.10 \text{ USD}$$

- (a) Calculate the EUR/INR exchange rate.
 (b) If an importer needs €250,000, compute the INR amount payable.

9. The following exchange rates are observed:

$$1 \text{ USD} = \text{Rs.}82.80, \quad 1 \text{ GBP} = 1.30 \text{ USD}, \quad 1 \text{ GBP} = \text{Rs.}108.50$$

- (a) Check whether triangular arbitrage opportunity exists.
 (b) If arbitrage exists, calculate the profit on an initial investment of Rs. 1,000,000.

10. An Indian exporter receives USD 500,000 after 3 months.

- Current spot rate: Rs. 82.40/USD
- Expected future spot rate: Rs. 83.10/USD

- (a) Compute the INR value receivable today.
 (b) Compute the expected INR value after 3 months.
 (c) Determine the gain or loss due to exchange rate movement.